LCLE: a web portal for lncRNA network analysis in liver cancer

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Abstract: Most of the currently available co-expression network analysis method only can capture linear correlation among genes; however, ignore the non-linear dependent correlations. Accurately and easily getting the distance values among genes are of significant importance in clustering genes which are shared in the same biological functions. We developed an online tool, lncRNA explorer (LCLE), which is able to systematically analyse gene expression data to identify more comprehensive relationships among lncRNAs and protein-coding genes (PCGs) from five different distances metrics. Our simulation results demonstrated that the selection of an appropriate distance method could help to identify novel important genes from networks. LCLE allows users to visualise figures, and download tables analysed from publically available RNAseq data such as TCGA and GTEx or upload their own data for analysis.

AQ1: Please supply expansion for the highlighted acronyms.

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Overall, our web portal will benefit for biologists or clinicians without programming background in identifying novel co-regulation relations for lncRNAs and PCGs.

Keywords: adjacency matrix; network analysis; correlation; non-coding RNA; cancer.

AQ2: Please increase keywords of no more than 10 phrases.

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1 Introduction

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Correlation networks are widely used in system biology applications such as estimating the co-expression patterns among genes, clustering genes shared in the same pathway, and selecting genes related to certain phenotypic conditions. In order to obtain optimal results, the key step is to fine-tune the distance parameters among genes. While many previous studies have developed methodologies for correlation network, no comprehensive studies comparatively investigated the outcome performance if choosing different distance metrics. Also, there are no user-friendly and reliable tools available so that users can have more options in choosing a distance metric based on their data distribution.

In the network analysis, determining which distance to use is probably one of the most difficult decisions (de Siqueira Santos et al., 2013). Providing more options to calculate the adjacency matrix of a network is increasingly important in detecting novel gene-gene and gene-traits associations. Especially the expression of long non-coding genes (lncRNA) is generally lower than protein-coding genes (PCGs), their dependent correlations are highly likely to be ignored using traditional linear correlation method such as Pearson correlation.

Also, another gap is that currently available weighted co-expression network methods such as WGCNA (Langfelder and Horvath, 2008) only has the power to detect linear correlation among genes due to the limited distance calculation options for the adjacency matrix. Our online liver cancer lncRNA explorer (LCLE) is a comprehensive collection of distance functions for performing various weighted or un-weighted correlation network analysis. In a weighted network, connections between genes using the similarity score between 0 and 1 as the weight to indicate the strength of correlations among genes. In an un-weighted network, users can set a distance cutoff value to determine if the genes are connected or unconnected based on if the genes with a correlation above the cutoff threshold.

Here, we provided the following five distance measurement including Pearson's correlation coefficient (Galton, 1889; Pearson, 1920) Spearman's correlation coefficient (Spearman, 1904), Kendall's correlation coefficient (Kendall, 1938), distance correlation (Székely et al., 2007), and maximal information coefficient (MIC) (Reshef et al., 2011) in our LCLE web portal. We mentioned 'correlation' here, which is an inverse of distance. This work seeks to identify a dependency between protein-coding and non-coding genes (i.e., lncRNAs, pseudogenes) from liver cancer RNAseq data. We discussed sample size, local dependence and linear/non-linear and also non-functional relationships.

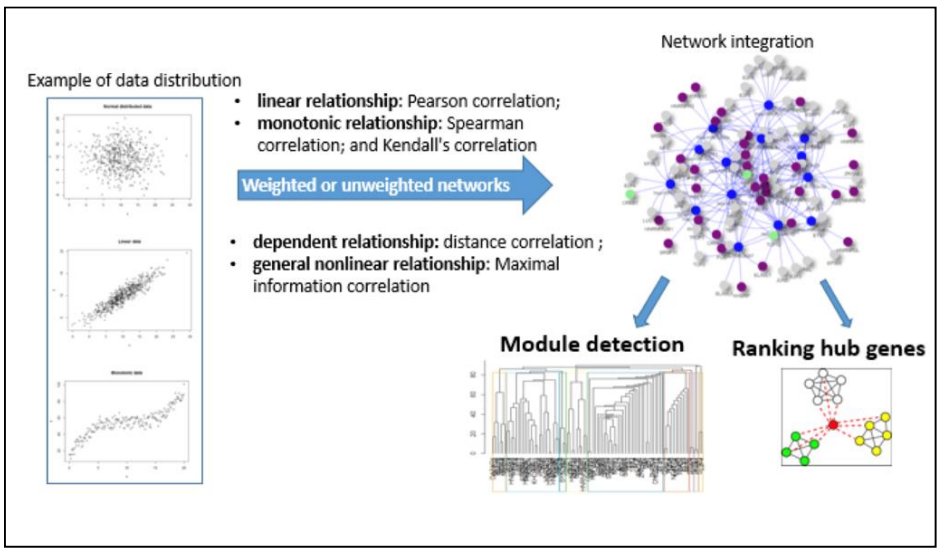
2 Materials and methods

2.1 Project design and data resource

The framework of the project design can be found in Figure 1. To show the analysis capacity of a various data matrix, we applied our data on RNAseq data downloaded from GTEx (n = 110) normal liver samples and TCGA (n = 369) tumour and (n = 50) normal samples as well as clinical data. All these coding and non-coding gene expression data are processed by the same Toil pipeline (Vivian et al., 2017) developed by UCSC. Given that the non-coding transcript expression of RNAseq data contains many types of

ncRNAs, we mainly group them into three big categories: PCGs, lncRNAs and pseudogenes. Specifically, “lincRNA”, “antisense”, “sense intronic”, “sense overlapping”, “processed transcript” are grouped into lncRNAs; “pseudogene”, “processed pseudogene”, “transcribed processed pseudogene”, “unitary pseudogene”, “polymorphic pseudogene” are grouped into pseudogenes. To show liver cancer specific non-coding genes, we also performed differential expression analysis by R package limma (Ritchie et al., 2015).

Figure 1 Project workflow (see online version for colours)



2.2 Distance method description and comparison

The Pearson’s correlation, Spearman’s rank correlation and Kendall τ rank correlation are the most popular correlation measurement to capture linear relationship between two variables X and Y. In details, the coefficient value of Pearson’s correlation [0, 1] implies positive linear correlations; while [–1,0] implies negative linear correlations (Speed, 2011). Spearman’s rank correlation coefficient and Kendall’s τ rank correlation coefficient measure how well the monotonic relationship between variables, and the assumption of linearity does not required. If there are no repeated data values, a perfect Spearman correlation of +1 or –1 occurs when each of the variables is a perfect monotone function of the other. While Kendall τ rank correlation coefficient is used to measure the ordinal association between two measured quantities. The Kendall correlation between two variables will be high when observations have a similar rank, and low when observations have a dissimilar rank. It is common to regard these two rank correlation coefficients as alternatives to Pearson’s coefficient, used either to reduce the amount of calculation or to make the coefficient less sensitive to non-normality in distributions (Ding and Li, 2013; Kendall, 1948).

Distance correlation was introduced in 2005 by Gábor J. Székely to capture the dependence relationship between two variables (Székely et al., 2007). Distance correlation satisfies $0 \leq \mathrm{dCor}(X, Y) \leq 1$. And $\mathrm{dCor}(X, Y) = 0$ if and only if X and Y are

independent. Thus, distance correlation measures both linear and nonlinear association between two random variables. MIC is based on naïve Mutual information computed using a data dependent binning scheme. The main idea of MIC is to discretise the data onto many different two-dimensional grids and calculate normalised scores that represents the mutual information of the two variables on each grid. The scores are normalised to ensure a fair comparison between different grids and vary between 0 (uncorrelated) and 1 (high correlations). MIC captures a wide range of associations both functional and not, and for functional relationships provides a score that roughly equals the Pearson's coefficient of determination (R^2) of the data relative to the regression function.

The choice of which method to use depends on which behaviour seems most reasonable for your purposes. If you are okay with discarding unrelated, dependent variables, then the Pearson's coefficient is fine. But if you truly want unrelated, dependent variable to be considered, then the nonlinear method is probably best. The following table list several measurements to the performance of methods. Generality means that with sufficient sample sie the statistic should capture a wide range of interesting associations, not limited to specific function types (such as linear, exponential, or periodic), or even to all functional relationships. Equitability means that the statistic should give similar scores to equally noisy relationships of different types.

2.3 Development of the correlation network for liver cancer RNAseq data

Detecting patterns across different distance conditions is becoming an important task. For example, it is interested in identifying gene modules and what's the difference among these networks. Data on gene networks can be acquired for multiple distance measurement in our LCLE web browser. In a weighted correlation network, each node represents a gene (feature), and links represent correlations between the genes. In an unsigned co-expression network, modules correspond to clusters of genes with high absolute correlations. Specifically, we used $a_{ij} = |\mathrm{cor}(x_i, x_j)|$ to represent the adjacency of a weighted network. Optionally, in an unweighted network, users can set a correlation cut-off to determine if the genes should be connected or not connected. The connectivity score is used to measure the connection strengths for each gene with the other network genes. Then we performed gene modules analysis to cluster genes of highly interconnected using Hierarchical cluster method.

3 Results

3.1 Analysis of simulations for distance measurement

As shown in Table 1, we evaluated the strengths and limitations of each distance method to help users determining which method to choose so that it marries well with their gene expression data. The choice of which method to use depends on which behaviour seems most reasonable for user's purposes. Pearson's coefficient is a better choice if users do not care about dependent relationship; while distance correlation is highly recommended if users are more interested in the unrelated, dependent relationship. However, generally, a large sample size should capture a wide range of interesting associations, no matter a linear or non-linear relationship.

Table 1 Comparative summary for five standard distance method

Methods	*MIC*	*Distance correlation*	*Kendall*	*Spearman*	*Pearson*
Application	Linear, Nonlinear	Linear, Nonlinear	Linear, Monotonic nonlinear	Linear, Monotonic nonlinear	Linear Data
Normalisation	Yes	Yes	Yes	Yes	Yes
Range	[0, 1]	[0, 1]	[–1, 1]	[–1, 1]	[–1, 1]
Generality	high	High	low	low	low
Equitability	Non-equitable (Kinney and Pearson, 2014)	Non-equitable (Kinney and Pearson, 2014)	Non-equitable (Kinney and Pearson, 2014)	Weakly-equitable (Ding and Li, 2013)	Non-equitable (Kinney and Pearson, 2014)
Symmetry	Yes	Yes	Yes	Yes	Yes

3.2 Performance evaluation for various distance on simulation data.

To show the performance of each distance metrics, we simulated six types of gene relationship to compare each method. Figure 2 illustrated six types of example relationships such as linear, monotonic, and cross relationships. The performance of statistic correlation of the five distance on six types of relationships were shown in Table 2. Pearson correlation performs best in linear or close linear distribution (monotonic type), but shows poor correlation coefficient in other dependent relations. Spearman and Kendall show a little better performance than Pearson, but still lower than what we expected on the nonlinear case. Distance correlation performs a little poorer than Pearson on linear relation, but simulation on nonlinear dependence examples are much better. MIC correlation shows a strong dependence on linear and nonlinear example except the cross one.

Figure 2 Illustration of simulated dataset with dependent relations. n = 500 data points generated for six relationship types with additive noise

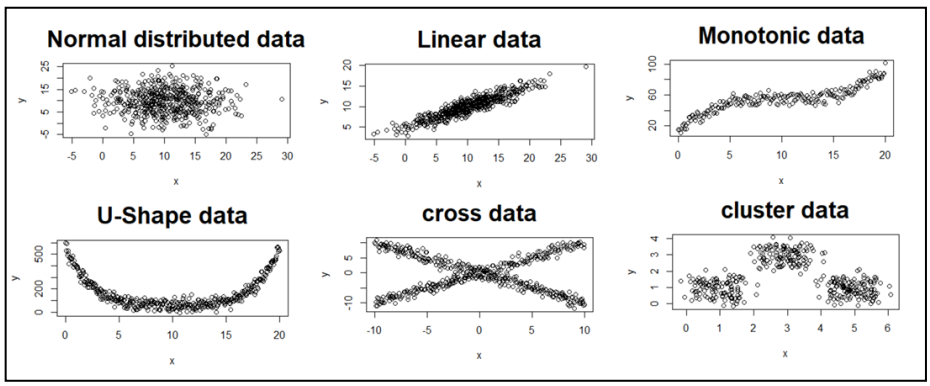


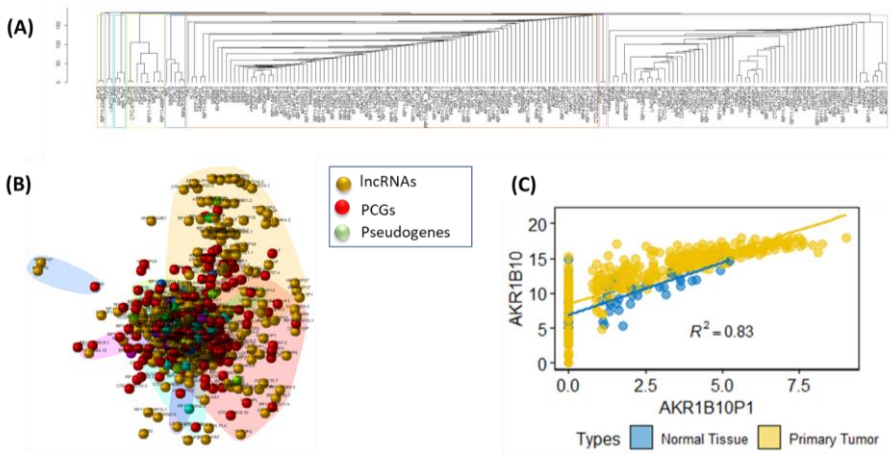
Table 2 Correlation performance of the five distance measures on six example distributions

Relationship type	*Pearson* $-1 \le r \le 1$	*Spearman* $-1 \le \rho \le 1$	*Kendall* $-1 \le t \le 1$	*Distance* $0 \le d \le 1$	*MIC* $0 \le r \le 1$
Normal distributed	–0.0412	–0.0334	–0.0228	0.0643	0.1597
Linear	0.9204	0.9093	0.7396	0.8905	0.7445
monotonic	0.8799	0.8762	0.7233	0.8523	0.7045
U–shaped	–0.0099	–0.0113	–0.0080	0.4387	0.7889
cross	–0.0581	–0.0569	–0.0458	0.2425	0.4706
Clustering	0.2963	0.3826	0.3806	0.3458	0.9535

3.3 Application of lncRNAs and PCGs co-expression network for liver cancer

To illustrate the applications of the five distance methods in retrieving relevant coding and non-coding genes regulatory relationships, we applied them to an RNA-seq data from liver cancer. Figure 3 illustrated the visualisation example of network cluster analysis based on Pearson correlation distance. Figure 3(A) and (B) showed the clusters by the hierarchical tree structure and network modularity analysis respectively. Figure 3(C) showed an example of a linear relationship between two genes, which is easily detected by Pearson correlation method.

Figure 3 Illustration of gene clustering based on Pearson correlation method (see online version for colours)



3.4 Web portal visualisation

We deployed all these functions in the LCLE (https://datasciences123.shinyapps.io/LCLE), we have provided network visualisation for different distance metrics for Liver cancer RNAseq data we also allow the users to perform a network analysis for their own data with a restricted 2000 genes due to the hardware limitations. Users can download the figures and statistic summary tables analysed by our tool.

4 Discussion

One major task in computational system biology is to understand the dependency among genes to model gene correlation networks. Correlation networks facilitate network-based gene screening methods that can be used to identify candidate biomarkers or therapeutic targets. However, the currently available network tools focus on a variety of fancy and comprehensive algorithms while ignored a basic fundamental step for the correlation distance calculation. Here, we provided users an interactively web portal so that users can have more options in genes distance measurement based on their data distribution.

Pearson's correlation is the most common method used to measure the distance between gene expressions, but if the gene pairs are linearly associated, the results will be optimal. For other types of association, such as non-linear or non-functional relationships, methods based on the rank correlation and information theory are more adequate than the Pearson's correlation but are less used in applications, most probably because of a lack of clear guidelines for their use. Moreover, given the same dataset, significant differences in conclusions may arise from the selection of distinct distance measures. Therefore, providing more options for distance selection is an appropriate method so that users can identify novel meaningful genes.

However, if the data contain a lot of technical artefacts or noise, the results will be very sensitive. Because of the unsupervised clustering in network analysis, it is important to detect f unwanted variations. While the methods development was motivated by gene expression data, the underlying data mining approach can be applied to a variety of different types of data. More importantly, not limited to network analysis, any clustering or classification problems of samples or genes involved with choosing an appropriate distance, because some distance metric will be sensible than others, depending on the properties of the given dataset.

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